

KIM SOOYEON

Undergraduate
Department of AI Convergence, Artificial Intelligence Major

PERSONAL DATA

- Birth: November 23, 2003. Seolleung-ro, Gangnam-gu, Seoul
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EDUCATION

Mar.2022 – **Sungshin Women's University**
Feb.2026 B.S. in AI Convergence, Major in Artificial Intelligence
Total GPA of 3.81/4.5 (Credits taken: 133/130)

RESEARCH INTEREST

- Human-AI Interaction
- Large Language Models (LLMs)
- On-Device / Applied AI Systems

INTERNSHIPS

Mar.2024 – **Computer Vision Lab Undergraduate Researcher**
Sep.2025 - Conducted weekly seminars including studying with the textbook “모두의 딥러닝 | 조 태호”, solving Kaggle problems, and presenting progress of ongoing projects.
- In the Undergraduate Research Participation Program (UROP), conducted research titled “Structured and Grounded Chain-of-Thought for Context-Aware Facial Emotion Recognition”, implementing a zero-shot VLM→LLM pipeline (BLIP-2/ExpansionNet + GPT-3.5) with structured and grounded Chain-of-Thought prompting, and improving emotion recognition reliability via late fusion, dynamic label selection, and calibration with rejection.

AWARDS AND HONORS

2025 **Student Invention Idea Contest & IP Mentoring Program**, Sungshin Women's University
- **Excellence Award (3rd Prize)**
- Proposed a patent-focused multi-agent analysis workflow and implemented automated noise filtering using a TF-IDF + Shannon Entropy scoring scheme with an LLM-assisted threshold recommendation.
- Performed LLM-based zero-shot technology classification on the filtered patents and generated structured outputs for downstream patent-map style analysis.

2025 **Academic Excellence Scholarship (Merit-based)**, Sungshin Women's University
- Awarded for Fall 2024 academic performance (disbursed Spring 2025)
- Awarded for Spring 2025 academic performance (disbursed Fall 2025)

2024 **Sungshin Women's University Student Competition — Carbon Footprint App**
- **Encouragement Award (Honorable Mention)**
- Developed a mobile app that tracks users' carbon footprints and provides personalized eco-friendly action recommendations.

PATENTS

Apr. 2026 **Automatic Patent Analysis Method Based on Patent Data Noise Removal**
(특허 데이터 노이즈 제거 기반 자동 특허 분석 방법)
- Korean Patent Application No. 10-2026-0066943, filed Apr 14, 2026
- Applicant: Sungshin Women's University Industry-Academic Cooperation Foundation
- Inventors: Oh Jang Min, Son Chai Eun, Kim Soo Yeon, Woo Hee Won
- Status: Pending (Examination Requested)

PROJECTS

- 2026 **LG Aimers 8th Cohort — EXAONE Model Compression Hackathon**
- Performed model compression on EXAONE-4.0-1.2B, optimizing the trade-off between model size, inference speed, and performance
 - Applied GPTQ post-training quantization and compared quantization schemes (W8A8, W4A16) to analyze trade-offs
 - Applied parameter-efficient fine-tuning (qLoRA) to maintain or improve model performance under a fixed vLLM-based inference environment
 - Submitted Hugging Face-compatible model weights and configurations for evaluation
- 2025 **Campus Patent Universiade (Korea) — Generative AI Patent Analysis**
- Selected as a finalist team (Top 6 teams in the track).
 - Conducted patent trend analysis on Generative AI applications (text, image/video, audio, and multimodal generation).
 - Built KIPRIS search queries with refined keyword filtering to reduce noise patents.
 - Performed topic discovery via clustering (HDBSCAN, K-Means, Spectral Clustering) and extracted representative keywords using KeyBERT, KRWordRank, and NounRank.
 - Contributed to writing and structuring the patent-based technology forecasting report.
- 2025 **LG AIMERS 6th Final Hackathon (Top 4%)**
- Built a soft-voting ensemble model based on CatBoost, LightGBM, and XGBoost to predict the probability of pregnancy success using treatment data from infertile patients.
- 2025 **Hanium Dream-Up: On-Device Medical Interview Chatbot Based on DeepSeek**
- Fine-tuned the DeepSeek-R1-Distill-Qwen-7B model using qLoRA for medical interview dialogue purposes
 - Developed a medical interview chatbot website and deployed the quantized model on Jetson Nano for on-device inference execution
- 2025 **Real-Time Korean Sign Language (KSL) Recognition & Translation System**
- Developed a real-time KSL translation pipeline using OpenPose keypoint sequences and morpheme-level annotations from the AI Hub Korean Sign Language dataset.
 - Implemented preprocessing and alignment between keypoint frames and gloss/morpheme timestamps to build sequence-to-text training data.
 - Designed and evaluated models including CNN+LSTM, BiLSTM, and Transformer Encoder + CTC for continuous sign recognition and subtitle generation.
 - Improved robustness by handling inconsistent OpenPose JSON formats and addressing CTC decoding issues such as blank collapse.
- 2024 **BloomBuddy — Android Flower Encyclopedia App (Course Project)**
- Developed an Android application that provides flower information including meanings, care tips (watering/light), and seasonal guidance using Kotlin.
 - Designed structured data models (e.g., `Flower` data class) and implemented a list-based UI (`activity_flower_list`) for browsing flower entries.
 - Explored database integration using SQLite/Firebase for efficient data storage and retrieval.
- 2023 **Database-Driven Music Streaming Service Design & Implementation**
- Designed a relational database schema for a music streaming service with core entities such as users, songs, artists, albums, and playlists.
 - Implemented SQL queries for key features including song search, playlist management, and play-history logging.
 - Applied normalization to reduce redundancy and used JOIN-based retrieval logic for service-level data access.

TRAINING & CERTIFICATIONS

2026	TOEIC - Score: 810
2026	ADsP (Advanced Data Analytics Semi-Professional) - Certified by Korea Data Agency
2025	Patent Investigation & Analysis Practical Training Program (30 hours), KISTA - Completed a university-tailored hands-on patent analysis program (Jul–Aug 2025). - Covered prior-art search, patent landscape analysis, and patent strategy report writing. - Produced and submitted a patent strategy report as part of the Campus Patent Universiade preparation.
2025	SK AI Dream Camp (Intermediate), mySUNI - Completed hands-on training covering end-to-end ML workflows from data analysis to model development. - Developed industry-focused ML projects: photo process indicator-based package outcome prediction, Btv viewing-data recommendation modeling, and SK ecoplant rebar unit price forecasting.